

# Evoking Harmony via Convolution

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**Abstract.** I show how to evoke the pitch-class content of a chord from an arbitrary source by convolving it with a short impulse response whose grains lie on those pitch-classes in every octave. A Csound implementation (`chord_convolver`) builds one joint IR with frequency-dependent grain length, equal-energy normalization across octaves, inverse A-weighting, a dry Dirac, and a single `ftconv`. I contrast the effect with a linear comb and a generic constant- $Q$  bank, and demonstrate it on a twilight field recording alongside Château de Lagarde.

**Keywords:** csound, harmony, convolution, constant- $Q$ , field recording

## 1 Introduction

As a boy crossing the desert at night I heard melodies and chords in the wind outside my father’s car. This paper reports Csound [1] experiments that evoke such harmonies from varied sources by convolving a signal with an impulse response whose grains occupy the pitch-classes of a chosen chord on every octave. Materials are in my GitHub repository [2].

The effect sits at the boundary of timbre and tonality, against a long-standing disjunction between “non-tonal” electroacoustic practice (acousmata) and tonal practice (pitch relations abstracted from actual sounds). That split goes back to the 1950s–1960s mix of technological utopianism and musical serialism. A constant- $Q$  bank *centered on those pitch-classes* is closely related; a generic constant- $Q$  tiling is only loosely related; a comb filter is less related still (periodic in linear Hz); Gabor/Morlet cells not centered on pitch-classes are a poor fit.

## 2 Mathematics

A *grain* is one windowed sinusoid (pitch-class  $c$ , octave  $n$ ). Its frequency response is a *kernel*  $H_{c,n}$ . The *impulse response* is the sum of the loudness-balanced grains plus a Dirac, then jointly  $\ell_2$ -normalized. For a chord  $C$  those kernels form an octave-related scaled family:

$$H_C(f) = \sum_{c \in C} \sum_n H_{c,n}(f), \quad H_{c,n}(f) \simeq H_c(2^{-n}f).$$

With  $u = \log_2 f$ , octave shift is unit translation and  $H_C(u + 1) \simeq H_C(u)$ : the chord is the unit cell of a log-periodic filter. Grain length

$$T(f) = \min(T, T \cdot 440/f)$$

is constant- $Q$  above 440 Hz; below the pivot, duration is capped at  $T$  [3].

### 3 Method

`chord_convolver` selects MIDI notes whose pitch-classes are in  $C$  (reduced modulo 12; negatives unused) and assigns each partial  $f$  a causal half-cosine of length  $T(f)$ , with  $T$  clamped to  $[0.01, 1.5]$  s (starts at 1, fades to 0). Grains are levelled by octave, a Dirac is added at sample 0, the IR is jointly  $\ell_2$ -normalized, and `ftconv` [4] is applied once. Full signatures and code are in the repository [2].

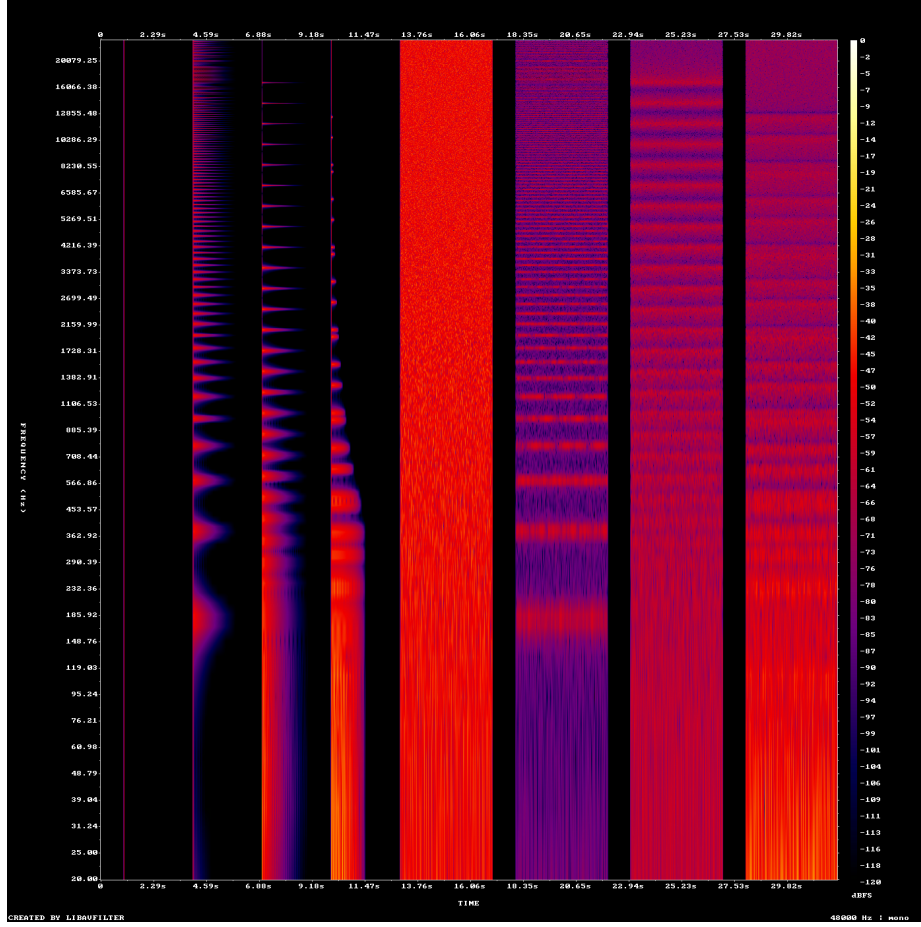
Both the causal window and the cap on  $T(f)$  serve auditory fusion. Since  $Y(f) = H(f)X(f)$  vanishes wherever the source has no energy, the lattice can only expose pitch content the source already contains; what makes that content read as latent in the source rather than as an added voice is keeping the harmonic response inside the source’s own events, since energy persisting after the source falls silent would be a cue for a separate resonating object. On a 50 ms impulse response driven by a 3 Hz impulse train, this design leaves under 3% of each period’s energy outside the interval the dry source occupies, against a third or more for an uncapped or symmetrically windowed grain.

Levelling the octaves takes two stages. Considering a source as broadband, the output power contributed by partial  $k$  is  $S_0 E_k$  by Parseval’s theorem, so equal presence requires equal grain *energy*, giving  $A(f) \propto T(f)^{-1/2}$ ; a constant factor holds total grain energy fixed so the impulse and Dirac gains keep their meaning. Loudness tracks power rather than spectral density because every grain occupies less than one auditory filter—at most 0.74 of an ERB [5]. Normalizing the window integral instead, which fixes peak transfer gain, tilts the lattice bright by some 8 dB, because the grain bandwidth  $\approx 1/T(f)$  widens as the window shortens. The second stage is inverse IEC A-weighting (shelf at 150 Hz, boost capped at +6 dB), which turns equal power into approximately equal loudness. Equalizing critical bands rather than partials, which attenuates the crowded low octaves by 4 to 6 dB, is available as an option but is not used here (compensation 1; no ERB correction).

A chord-restricted constant- $Q$  bank is a near neighbour (same lattice, ongoing IIR filtering). A generic three-semitone constant- $Q$  tiling lacks chord identity. A comb with delay  $1/f_0$  has equal linear-Hz spacing.

### 4 Results

Figure 1 compares click and noise under dry, comb ( $f_0 = 200$  Hz), sparse CQT (every three semitones), and evoke with PCs 0, 4, 7, 11 (CM7), loudness-matched to within 1.1 dB A-weighted within each source group. Comb is linear-periodic;



**Fig. 1.** Log-frequency spectrogram (left to right): dry click; click comb; click CQT; click evoke CM7; then the same four treatments of noise.

CQT is log-periodic but not chordal; the convolver shows the CM7 lattice, articulated to the top of the audible range under equal-energy normalization.

A twilight walk alongside Château de Lagarde recorded with a Tascam DR-40 is processed under score control of  $T$ , wet/dry, compensation type (here always 1, equal energy), and pitch-classes (MIDI key numbers, reduced modulo 12) [6]. Moments include a chord progression within voices, bells with a subtle effect, bells again with a stronger effect and voices following, and “*Bonjour*” with voices following.

## 5 Discussion

Evoking harmony is chord-specific and log-periodic, unlike a comb, and uses finite causal grains plus a dry Dirac in one impulse response rather than a steady filterbank. It is most appealing when harmonic sensation remains subtle relative to the source; wetter settings yield a clearer shadow or ring, or even more abstract harmonic and rhythmic textures. To see more: supplementary figures and audio on GitHub [2] include audio and orchestra code.

*Note.* This abbreviated version was edited for length with assistance from generative AI tools; the author reviewed the text and retains responsibility for all technical content.

## References

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